Coding & Solution

Date: 15 July 2026

Team ID: XXXXXX

Project Name: Human Development Index (HDI) Prediction using Machine Learning

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Solution Summary

Field	Details
Repository Link / URL	https://github.com/golidevisriprasad/smartbrige
Programming Language(s)	Python, HTML5, CSS3, JavaScript, SQL
Framework(s) Used	Flask, Scikit-learn, Pandas, NumPy, Bootstrap
Key Features Implemented	• User Authentication (Login)• HDI Dataset Upload & Management• Data Preprocessing• Machine Learning-based HDI Prediction• Interactive Dashboard & Charts• Country Comparison• Prediction History• Download Reports (PDF/CSV)• Responsive Web Interface
Pending / Incomplete Features	• Live integration with UNDP API (Future Work)• Multi-language support• Mobile Application Version• AI-based policy recommendation module
Setup / Run Instructions	**1.** Install Python 3.10 or above.**2.** Clone the GitHub repository.**3.** Install dependencies using `pip install -r requirements.txt`.**4.** Configure the MySQL database.**5.** Run the application using `python app.py`.**6.** Open `http://localhost:5000` in your browser.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions/classes	**Yes**
2	Meaningful variable and function names are used	**Yes**
3	Code includes comments/documentation where necessary	**Yes**
4	Error handling is implemented for critical operations	**Yes**
5	The application runs without critical errors	**Yes**
6	Code is committed to a version control repository	**Yes** *(GitHub)*

Additional Notes / Comments

- The project follows a **modular architecture**, separating the frontend, backend, machine learning model, and database layers for improved maintainability.
- **PEP 8 coding standards** are followed for Python source code, while HTML, CSS, and JavaScript files use consistent formatting and naming conventions.
- The machine learning model is developed using **Scikit-learn**, with **Pandas** and **NumPy** for data preprocessing and **Matplotlib** for visualization.
- User inputs are validated before prediction, and exception handling is implemented to ensure reliable system performance.
- The application is version-controlled using **Git** and **GitHub**, enabling collaborative development and easy maintenance.
- The solution is scalable and can be extended with additional features such as cloud deployment, real-time data integration, and advanced analytics.

Project Folder Structure

```
HDI_Prediction_System/
│
├── app.py
├── requirements.txt
├── README.md
│
├── dataset/
│   └── hdi_dataset.csv
│
├── model/
```

```
│   ├── train_model.py
│   ├── predict_hdi.py
│   └── hdi_model.pkl
│
├── templates/
│   ├── index.html
│   ├── login.html
│   └── dashboard.html
│
├── static/
│   ├── css/
│   ├── js/
│   └── images/
│
├── database/
│   └── database.sql
│
├── utils/
│   ├── preprocess.py
│   ├── visualization.py
│   └── helper.py
│
└── reports/
    ├── prediction_history/
    └── exported_reports/
```

Coding & Solution Summary

The **Human Development Index (HDI) Prediction System** is implemented using **Python, Flask, Scikit-learn, HTML, CSS, JavaScript, and MySQL**. The application provides secure user access, efficient data preprocessing, machine learning-based HDI prediction, interactive dashboards, and downloadable reports. The codebase is modular, well-documented, reusable, and follows software engineering best practices, making the system reliable, maintainable, and scalable for future enhancements.